Yaoyuan Zhou

My Website amiao1120@gmail.com GitHub

Education

Beijing Normal University *Sept 2023 – Jul 2027*
BS in Educational Technology (major) **GPA:3.7/4.0**
BS in Artificial Intelligence (minor)
Hong Kong Polytechnic University *Sept 2025 – Jan 2026*
Exchange Student - BSc(Hons) in Artificial Intelligence and Information Engineering

Awards

First-Class Merit-Based Scholarship**(Top 5%)**, Second-Class Jingshi Scholarship **(Top 20%)**,
Mitacs Global Research Internship Award **(200 participants nationwide)**
The NTU National Institute of Education (NIE) Singapore Undergraduate Online Scholarship

Research Interest

My research explores **AI-integrated instructional design and AI literacy**, focusing on how learners are introduced to generative AI and develop productive and self-directed ways of engaging with it in authentic learning contexts. **From the standpoint of Education, leveraging the potential of Technology in pedagogical practices to explore pathways towards harmonious coexistence between humanity and the world.**

Publications

Journal Articles

1. An Active Instructional Approach Based on SAMR Framework to Assist Undergraduate Freshmen in Integrating AIGC into Learning: An Exploratory
The Internet and Higher Education**(SSCI Q1, Third Author)** Link
Juan Wu*, Jingwen Pan, **Yaoyuan Zhou**, Mengyu Liu, Yanling Li, Ronghuai Huang,

2. Exploring the Second-Level Digital Divide in Elementary Science Classrooms: A Video-Based Comparative Analysis
International Journal of Science and Mathematics Education**(SSCI Q1, Second Author)** Link
Yuze He, **Yaoyuan Zhou**, Juan Lai, Fengrui Xie, Yonghe Zheng, Jie Yang*

Conference Papers

3. Exploring the Pattern and Influence Factors in the Correlation Between Learning Engagement and Academic Achievement in Online Learning
Global Chinese Conference on Computers in Education, GCCCE 2026 **(First Author)** Link
Yaoyuan Zhou, Yuhan Gai, Qingyang Liu, Jiamin Ren

4. Power of Query——Analysis of Critical Thinking Trajectories Based GAI-Assisted Learning
Global Chinese Conference on Computers in Education, GCCCE 2025 **(First Author)** Link
Yaoyuan Zhou, Juan Wu*

5. A Typological Study of Learning Engagement and Academic Achievement in University Online Classes :An Epistemic Network Analysis Based on Bloom's Taxonomy
Research Association of Learning Science Conference, RALS 2025 **(First Author)**
Yaoyuan Zhou, Yuhan Gai, Qingyang Liu, Jiamin Ren

Manuscripts Under Review

6. Exploring the Effects of Generative AI Prompting Instruction on Undergraduate Students' Creative Problem Solving: From Learning to Using in a Classroom-Based Social Experiment
Smart Learning Environment (**Third Author**)
Juan Wu*, Jingwen Pan, **Yaoyuan Zhou**, Mengyu Liu, Ronghuai Huang

7. Guide, Don't Distract: Inference-Time Control for Non-Intrusive AI Museum Guides
ACM Conference on Human Factors in Computing Systems (CHI)
Vanessa Figueiredo , Hilda Cruz Sanguino , Ianca Silva, **Yaoyuan Zhou**,

8. Pathways of academically productive talk: An Epistemic Network Analysis of teacher discourse development across school contexts
International Conference of the Learning Sciences (**Second Author**)
Pengjin Wang , **Yaoyuan Zhou**, Liru Hu, Gaowei Chen

Project

Enhancing Education: A Human-Centred Generative AI Pipeline for Academic Excellence — ***Research Assistant*** *Jul 2026 - Oct 2026*
Mitacs Globalink Research Internship Program (University of Regina) - Canada

- Conducted mixed-methods research on visitor preferences for conversational AI museum companions, identifying design implications for agent initiative, and contributing to an ACM CHI 2027 manuscript.
- Analyzed teacher interviews using Cognitive Work Analysis and translated classroom scaffolding practices into design principles for conversational AI, contributing to an ACM IDC 2027 manuscript.

Exploring the Effects of GAI Prompting on Undergraduate Creative Problem Solving: A Task-Driven Experimental Study — ***Person in Charge*** *May 2024 - Dec 2025*
Municipal-level scientific research project

- Led the end-to-end design and implementation of a classroom AI-agent platform supporting real-time student–GAI interaction.
- Designed a prompt-engineering curriculum and analyzed students' chatbot dialogues using Epistemic Network Analysis (ENA) to examine their critical and creative thinking.

The scientific and engineering education discipline community of Beijing Normal University — ***Project Intern*** *Oct 2023 – Jun 2025*
Research Institute of Science Education

- Designed a 12-hour aviation-themed STEM curriculum grounded in Project-Based Learning (PBL), developing hands-on learning tasks and co-authoring the accompanying textbook materials.
- Collaborated with 20+ frontline STEAM educators, serving between the research team and teachers to translate classroom feedback into iterative curriculum revisions and coordinate textbook development.

Work Experience

International Practicum Student — *Jan 2027 – Feb 2027*
National Institute of Education International Practicum, NTU (Inbound) – Singapore

- Selected for a practicum at Hwa Chong Institution, teaching Mathematics and lower-secondary science while observing Singapore's classroom practices.

Information and Communication Technology Project Intern — *Feb 2025 – Aug 2025*
UNESCO International Research and Training Centre (INRULED)

- Facilitated the joint SEAMEO STEM-ED and UNESCO INRULED "STEM Project" by conducting background research on rural STEM education, drafting cooperation frameworks and evaluation criteria.
- Assisted about 5 large-scale international events, managing concept notes, round-table session design, foreign expert liaison, and on-site multimedia in high-pressure, multilingual settings.
- Conducted field surveys across Chongqing, Jiangsu, and other regions to assess the current implementation status of ICT education.

AI Introduction Online & Offline Teacher Volunteer — *Jan 2024 - Jan 2025*
HER Digital Future Bootcamp by United Nations Development Program (UNDP)

- Served as an AI literacy instructor at the 2024 HER Digital Future Camp, delivered 20 hours of instruction to primary school students and promoting girls' participation in STEM through hybrid learning experiences.
- Developed introductory AI lessons based on human–AI interaction, fostering students' critical thinking, problem-solving abilities, and early self-regulated learning skills.

Special Skills

Languages: English(IELTS 7.0),Chinese(Native Speaker), Cantonese(Beginner)
AI Capacity: PyTorch, Transformers, scikit-learn, pandas, NumPy; BERT and LLM-based text classification
Research Method: ENA, process mining, lag sequential analysis, systematic review and meta-analysis